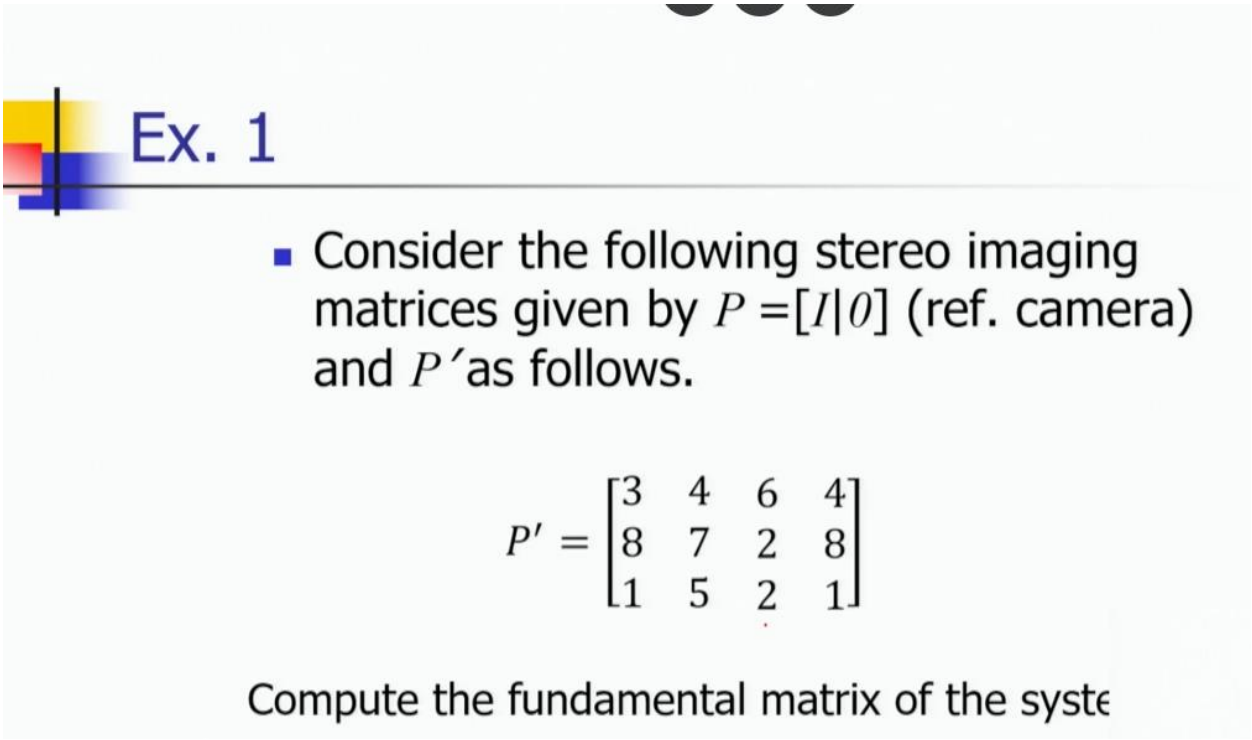


We're given the stereo cameras:

- $P = [I \mid 0]$
- $P' = \begin{bmatrix} 3 & 4 & 6 & 4 \\ 8 & 7 & 2 & 8 \\ 1 & 5 & 2 & 1 \end{bmatrix}$

Step 1: Key formula

For a canonical camera $P = [I \mid 0]$, the fundamental matrix is:

$$F = [e']_\times M'$$

where:

- M' = first 3 columns of P'
- e' = epipole in the second image = last column of P'
- $[e']_\times$ = skew-symmetric matrix of e'

Step 2: Extract components

$$M' = \begin{bmatrix} 3 & 4 & 6 \\ 8 & 7 & 2 \\ 1 & 5 & 2 \end{bmatrix}, \quad e' = \begin{bmatrix} 4 \\ 8 \\ 1 \end{bmatrix}$$

Step 3: Skew-symmetric matrix of e'

$$[e']_\times = \begin{bmatrix} 0 & -1 & 8 \\ 1 & 0 & -4 \\ -8 & 4 & 0 \end{bmatrix}$$

Step 4: Compute $F = [e']_\times M'$

$$F = \begin{bmatrix} 0 & -1 & 8 \\ 1 & 0 & -4 \\ -8 & 4 & 0 \end{bmatrix} \begin{bmatrix} 3 & 4 & 6 \\ 8 & 7 & 2 \\ 1 & 5 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 33 & 14 \\ -1 & -16 & -2 \\ 8 & -4 & -40 \end{bmatrix}$$

✅ **Final Answer (up to scale)**

$$\boxed{F = \begin{bmatrix} 0 & 33 & 14 \\ -1 & -16 & -2 \\ 8 & -4 & -40 \end{bmatrix}}$$

(Any scalar multiple of this matrix is also valid.)

📌 **Definition**

A camera is called **canonical** if its projection matrix is:

$$P = [I \mid 0]$$

where:

- I = 3×3 identity matrix
- 0 = zero translation vector

💡 **What it means intuitively**

A canonical camera:

- Is located at the **origin** $(0, 0, 0)$
- Has **no rotation** (aligned with world axes)
- Has **no translation**
- Acts like a "reference" camera

So it's basically the **simplest possible camera setup**.

📌 Why we directly use M'

We split:

$$P' = [M' \mid e']$$

Then the formula becomes:

$$F = [e']_\times M'$$

No inverse appears because:

- M' already maps 3D world points into the second image
- The first camera is identity → no back-and-forth transformation needed

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🧠 Intuition

Think of it like this:

- Normally:
  "Go from camera 1 → world → camera 2"
  → requires inverses
- Canonical case:
  "Camera 1 **is** the world"
  → no inverse needed

⚠️ When WOULD you use M^{-1}?

You'd need an inverse if:

- The first camera is **not** $[I \mid 0]$
- You're converting between different coordinate frames
- You're computing homographies or essential matrices in general form

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✅ Final takeaway

No M^{-1} is used because:

✔ The first camera is canonical
✔ The coordinate system is already aligned
✔ The formula simplifies to:

$$F = [e']_\times M'$$